

(TR-104) PYTHON WITH AI/ML

Training Day 65 Report

Date: 10 April 2026

Topic Covered: Instagram Trend Sources, Reel Script Generator & System Testing

The sixty-fifth day of training focused on exploring reliable sources for Instagram trend extraction, building an automated Instagram reel script generator, and performing website module testing. The session also included evaluation of scraping tools, API limitations, and fallback strategies for real-world robustness.

- Researched and evaluated multiple open-source solutions for Instagram trend extraction.
- Identified limitations of scraping-based methods due to Instagram's anti-bot restrictions and unstable page structure.
- Evaluated instagram-scraper and found it outdated and incompatible with current platform changes.
- Analyzed Instaloader and Instagrapi, noting dependency on login credentials and potential account restriction risks.
- Studied Instagram Graph API and understood its reliability but also its dependency on Business Account setup and approval.
- Tested Apify platform and observed limitations in free-tier usage and scalability constraints.
- Explored Reddit (PRAW API) as an alternative trend source but faced API/network access limitations.
- Developed a Python-based Instagram Reel Script Generator using requests, BeautifulSoup, and JSON processing.
- Implemented Instagram hashtag-based trend fetching using web scraping with proper request headers.
- Added fallback mechanisms to ensure trend availability even when scraping fails.
- Designed structured prompt engineering for generating Instagram reel scripts using LLMs.

- Integrated multiple Ollama models (qwen2.5:3b, 1.5b, 0.5b) for improved response stability and fallback support.
- Implemented regex-based JSON parsing to safely extract structured outputs from LLM responses.
- Added fallback script generation logic to ensure consistent output generation in all cases.
- Stored generated scripts in structured JSON format for reuse and analysis.
- Tested website modules of Mann Bros Leasing Dashboard including Inventory, User Management, and Recurring Check modules.
- Identified issue in Lease Info form where uploaded invoices are not opening correctly.
- Reviewed poster content and suggested multiple grammar and consistency improvements for better presentation quality.
- Strengthened understanding of web scraping challenges, fallback system design, structured prompting, and real-world system testing.
- **GitHub Link:**
<https://github.com/JaspinderKaurWalia26/trend>
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/instagram-reel-script-generator-using-python-and-ollama-874a25465e98>